4 **(a)** **(i)** The charge is positive as it is attracted to the negative plate (or repelled by the positive plate).

(ii) Since angle is 45°,

$F_E = mg$

$= (0.25 \times 10^{-3})(9.81)$

$= 2.4525 \times 10^{-3} = 2.5 \times 10^{-3}$ N

(b) $\tan\theta = \dfrac{F_E}{mg} = \dfrac{Q\,\Delta V}{d\,mg}$

graph of $\tan\theta$ against $\dfrac{1}{d}$ gives straight line through origin with gradient $= \dfrac{Q\,\Delta V}{mg}$

From graph, gradient $= \dfrac{150.0}{40.0}$

Hence $\dfrac{Q\,\Delta V}{mg} = \dfrac{150.0}{40.0}$

$Q = \dfrac{150.0}{40.0}\left(\dfrac{(0.25\times10^{-3})(9.81)}{12}\right)$

$= 7.6641\times10^{-4} = 7.7\times10^{-4}$ C

(c) When equilibrium is reached, $V_1 = V_2$

$\dfrac{Q_1}{C_1} = \dfrac{Q_2}{C_2}$

$Q_1 = \dfrac{Q_2}{C_2}C_1 = \dfrac{Q_2}{10} \times 5.0 = 0.50\ Q_2$ ..... (1)

Charges are conserved,

$Q_1 + Q_2 = C_1V_1 + C_2V_2 = (5.0\times10^{-12})(12.0) + (10.0\times10^{-12})(6.0) = 1.2\times10^{-10}$ C ..... (2)

Solving (1) and (2):

$Q_1 = 4.0 \times 10^{-11}$ C

$Q_2 = 8.0 \times 10^{-11}$ C